Pravallika Peddi

AI/ML Engineer

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AI/ML Engineer with 5+ years of experience building scalable, high-performance AI systems across robotics, distributed LLM infrastructure, and cloud-native ML platforms at Alphabet, NVIDIA, and Accenture. Specializes in large-scale distributed training (8K+ GPUs), reinforcement learning for robotic systems, and GPU-optimized deep learning architectures. My work spans end-to-end MLOps, cloud-native deployments on GCP and AWS, and production-grade inference systems designed for scale, reliability, and measurable performance gains. Passionate about solving complex engineering challenges where AI, systems performance, and real-world impact intersect.

EXPERIENCE

MACHINE LEARNING ENGINEER — **DEC 2024 – PRESENT**

ALPHABET — **MOUNTAIN VIEW, CA**

- Engineered learning-based perception and motion planning pipelines for industrial robotic systems, improving object detection accuracy by 28% and reducing task cycle time by 35% across manufacturing workflows.
- Built simulation-to-real transfer framework using synthetic data and reinforcement learning, increasing deployment reliability by 40% and decreasing physical robot testing costs by 30%.
- Designed scalable robot workflow orchestration platform integrating vision models and sensor fusion, improving automation throughput by 32% and reducing manual programming effort by 45%.
- Developed real-time perception systems using Python and C++, leveraging TensorFlow, PyTorch, and JAX for multimodal learning, pose estimation, segmentation, and adaptive robotic manipulation tasks.
- Implemented reinforcement learning and imitation learning algorithms for dynamic motion planning, combining model-based control, visual servoing, and sensor fusion in ROS2-driven robotic environments.
- Built high-fidelity simulation pipelines using Gazebo and NVIDIA Isaac Sim, enabling synthetic dataset generation, domain randomization, and robust sim-to-real generalization strategies.
- Collaborated cross-functionally with hardware, platform, and product teams, demonstrating strong communication, stakeholder alignment, agile execution, and technical leadership in complex AI-driven automation initiatives.
- Designed scalable MLOps pipelines with CI/CD automation, Docker, Kubernetes, Bazel, and gRPC-based microservices to support distributed model training, validation, and low-latency inference systems.
- Architected cloud-native training and inference workflows on GCP using Vertex AI, Kubernetes Engine, and distributed GPU/TPU clusters for scalable robotics model experimentation and deployment.
- Optimized end-to-end ML lifecycle on GCP integrating BigQuery, Dataflow, Pub/Sub, and monitoring pipelines, ensuring reliable data ingestion, experiment tracking, governance, and production-grade model observability.

SOFTWARE ENGINEER - MACHINE LEARNING — **OCT 2023 – NOV 2024**

NVIDIA — **NEW YORK, NY**

- Optimized multi-node distributed training pipelines across 8K+ GPU clusters, improving scaling efficiency by 22%, reducing communication latency by 18%, and accelerating large transformer training convergence.
- Re-architected collective communication strategies and kernel-level memory allocation, increasing cluster throughput by 27% and improving GPU utilization by 15% during large-scale LLM workloads.
- Enhanced inference runtime and mixed-precision optimization strategies, boosting tokens-per-second performance by 31% and reducing memory fragmentation by 17% across production-scale deployments.
- Engineered high-performance CUDA kernels and optimized GPU memory hierarchies, leveraging FP8 mixed precision, tensor parallelism, and warp-level scheduling for scalable transformer model acceleration.
- Designed distributed training architectures using PyTorch DDP, FSDP, DeepSpeed, and advanced model-parallel techniques, enabling stable multi-node scaling across heterogeneous GPU environments.
- Implemented communication optimizations using NCCL and topology-aware collectives, reducing synchronization overhead and enabling efficient gradient aggregation in extreme-scale data-parallel workloads.
- Built regression detection and benchmarking frameworks integrating ML observability, performance telemetry, and automated profiling pipelines using Nsight Systems and cluster-level monitoring tools.
- Collaborated cross-functionally with hardware, networking, and ML research teams, translating system-level bottlenecks into optimized AI infrastructure solutions using agile development methodologies.
- Developed production-grade MLOps workflows on AWS, integrating Kubernetes, EKS, S3, and distributed checkpointing pipelines to ensure fault-tolerant, elastic large-scale model training environments.
- Deployed scalable inference and monitoring pipelines on AWS, leveraging EC2 GPU instances, Auto Scaling, Terraform, and cloud-native observability stacks for resilient, enterprise AI infrastructure.

SOFTWARE ENGINEER **AUG 2019 – NOV 2022**
ACCENTURE **INDIA**

- Designed and deployed end-to-end machine learning pipelines for real-world AI prototypes, improving model accuracy by 28% and reducing experimentation cycle time by 35%.
- Led incubation of high-potential AI solutions into production-ready systems, optimizing inference performance by 40% and accelerating deployment timelines by 30% across cross-functional teams.
- Engineered scalable NLP and predictive analytics models using structured and unstructured datasets, increasing automation efficiency by 32% and reducing manual analysis efforts by 45%.
- Applied advanced Python, TensorFlow, PyTorch, Scikit-learn, and LLM frameworks to build classification, forecasting, and generative AI solutions with strong problem-solving and stakeholder collaboration skills.
- Implemented robust MLOps workflows including CI/CD, experiment tracking, feature engineering, drift detection, model explainability, and bias mitigation, demonstrating leadership, communication, and agile delivery expertise.
- Architected cloud-native ML solutions on AWS leveraging SageMaker, S3, Lambda, and Docker, enabling scalable training, real-time inference APIs, and secure model lifecycle management.
- Deployed production-grade AI systems on AWS using Kubernetes, EKS, MLflow, and Terraform, ensuring high availability, monitoring, cost optimization, and collaborative DevOps integration across distributed teams.

SKILLS

Programming & Core Languages: Python, C++, CUDA, SQL, Bash/Shell Scripting, Java, Scala, Go
Machine Learning & AI: Deep Learning, Generative AI, Large Language Models (LLMs), Transformers, Multimodal Learning, Reinforcement & Imitation Learning, Sim-to-Real Transfer, NLP, Computer Vision (Detection, Segmentation, Pose Estimation), Predictive Modeling, Foundation Model Optimization, ML Model Development, Data Structures, Data Modeling, Big Data
Distributed Training & GPU Systems: PyTorch (DDP, FSDP), DeepSpeed, TensorFlow, JAX, CUDA, NCCL, FP8 Mixed Precision, Tensor/Data/Model Parallelism, Multi-Node GPU Training, Inference Optimization, Memory & Kernel-Level Performance Tuning, Distributed Systems, Asynchronous Systems, Scalability, Performance Optimization
Robotics & Simulation: Perception Pipelines, Motion Planning, Sensor Fusion, ROS2, Gazebo, NVIDIA Isaac Sim, Synthetic Data Generation, Real-Time Robotic Systems
MLOps & AI Infrastructure: End-to-End ML Lifecycle, CI/CD for ML, MLflow, Model Registry, Drift Detection, AI Observability, A/B Testing, Automated Benchmarking, Kubernetes, Docker, Microservices, gRPC, DevOps Practices, Automated CI/CD, Infrastructure as Code, Containerization, Cloud Infrastructure, SDLC, TDD, Unit Testing, REST APIs, Serverless, Security, SRE
Cloud & Data Systems: GCP (Vertex AI, GKE, BigQuery, Dataflow, Pub/Sub), AWS (SageMaker, EKS, EC2 GPU, S3), Terraform, Distributed Data Pipelines, Batch & Streaming Systems, Cloud Computing, PaaS, ECS, Kafka, SQS, SNS, Apache Airflow, ETL Data Processing, Data Platform
Databases: Relational Databases, PostgreSQL, MySQL, SQL Server, NoSQL, DynamoDB, Elasticsearch, Redis, OLAP
Soft Skills: Technical Leadership, Cross-Functional Collaboration, Stakeholder Communication, Agile Execution, Systems Thinking, Performance Optimization Mindset, Ownership & Mentorship, Scrum, Sprint Planning, Code Review, Best Practices, Troubleshooting, Documentation, Analytical Skills, Critical Thinking, Continuous Improvement, Communication Skills

EDUCATION

University of Missouri **Kansas City, MO**
Master of Science in Computer Science

Hindustan College of engineering and Technology **India**
Bachelor of Electronics & Communication Engineering

CERTIFICATIONS

- Microsoft Certified: Azure Data Engineer Associate
- SnowPro® Core Certification
- Databricks Certified Data Engineer Associate